

TITLE PAGE

- Problem Statement ID – **SIH26155**
- Problem Statement Title- **AI-Driven Multi-Vendor Network Security Compliance Auditor**
- Theme- **Blockchain & Cybersecurity**
- PS Category- **Software**
- Team ID-
- Team Name - **C1TADΣL**



PROBLEM



- Multi-vendor networks suffer from extreme CLI syntax diversity.
- Legacy compliance tools, Traditional methods → manual, expensive, and vendor-locked.
- Small misconfigurations across devices cause major security breaches.
- Static parsers break on new hardware, firmware updates, and cloud setups.



NET-AUDIT

Our solution

- Ingest: bulk/live config upload, vendor auto-detected
- Normalize: extracts settings into one vendor-neutral schema
- Train on unknowns: admin labels new syntax once, remembered forever
- Evaluate: deterministic rule engine checks CIS/NIST/STIG/ISO
- Report: one PDF per device with findings + remediation CLI

Why It Scales

Without normalization

40 vendors
× 300 controls
= 12,000 pieces
of logic

With NetAudit

40 dictionaries
+ 300 controls
= 340 pieces
of logic

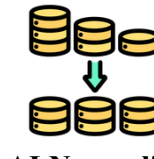
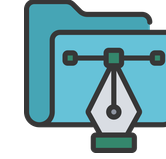
Adding vendor #6 costs one dictionary
— not 300 new checks

Innovation and Uniqueness

- AI only reads syntax verdicts are always rule-based, never guessed
- Errors become “needs review,” never a false compliance pass
- Coverage scales via data, not developer time
- Every finding shows its source line + confidence score
- Optional human-approved remediation layer closes the loop safely

Key Value Propositions

- Normalizes all vendors into one format logic written once
- New vendor = one new template, not a rebuild
- Training loop replaces developer work with one admin click
- Directly covers all 5 PS requirements
- Turns week-long manual audits into a single upload

Tech
Stack :**Frontend**
React.js + Tailwind CSS**Backend**
FastAPI (Python)**Network Data**
Netmiko / NAPALM**Database**
PostgreSQL + Hash-
Chaining**Deterministic Parsing**
TextFSM + NTC-Templates**AI Normalization**
SentenceTransformers
(all-MiniLM-L6-v2)**Vector Storage**
ChromaDB**Compliance Engine**
YAML Rules / OPA (Rego)**Reporting**
WeasyPrint / Jinja2**Deployment**
Docker Compose**CONFIG
INPUT**

File / livedevice

INGEST*FastAPI · Netmiko / NAPALM*

Flexible input - Single or bulk config uploads, plus live CLI pulls
Auto vendor detection - Identifies vendor and OS from syntax, no manual selection

NORMALIZE

Vendor-neutral output - Converts configs into one common JSON format, regardless of vendor
Easy to extend - New vendor = new template; the core engine never changes

*TTP / TextFSM · Pydantic***EVALUATE***OPA / Rego—deterministic*

Rule-based checks - Deterministic logic, not AI guesswork
Applicability first - Confirms a rule applies before scoring

PASS

FAIL

REVIEW

N/A

REPORT*Findings + evidence → remediation CLI → PDF*

Device identity - Serial number, model, and identity (live collection); marked "Unavailable" for file-only uploads
Evidence-backed findings - Every finding cites its source line and confidence score

*Jinja2 · WeasyPrint**Unknown / uncertain syntax*

AI SUGGESTS · RULES DECIDE

TRAIN ON UNKNOWN

UNKNOWN SYNTAX

MiniLM FINDS SIMILAR
EXAMPLESADMIN CONFIRMS
CATEGORY

MAPPING SAVED

Suggestions ≠ verdicts • Unresolved syntax → NEEDS REVIEW

Similarity search - MiniLM retrieves similar previously-labeled examples for unknown syntax**Human-in-the-loop** - Admin confirms only the category (never pass/fail); mapping is saved and auto-recognized from then on*MiniLM · ChromaDB · Training GUI**Reuse versioned mappings — no backend redeployment***OPTIONAL REMEDIATION****FINDING**Confirmed issue from
Evaluate**GENERATE FIX +
ROLLBACK****HUMAN APPROVAL**
Diff reviewed before apply**CONTROLLED APPLY**
One device at a time**RESOLVED**

Finding confirmed fixed

RE-AUDIT

Re-evaluates the finding

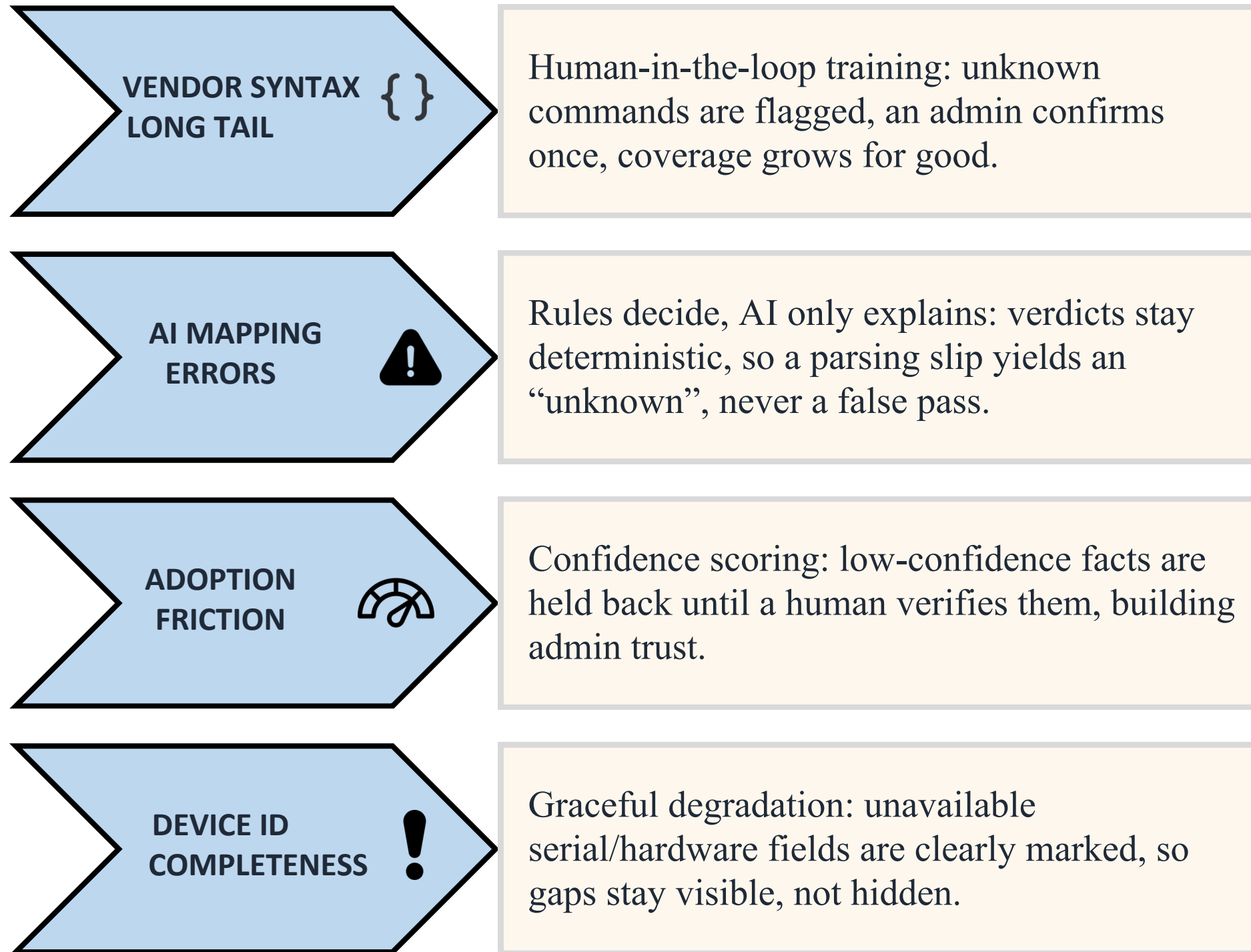
HEALTH CHECK

Confirms device still functioning

on failure → ROLLBACK / MANUAL RECOVERY*Git/Gitea · Netmiko/NAPALM · PostgreSQL*

- Safe by default
- Verified, not assumed
- Full audit trail

Challenges and Solutions



Technical & Architectural Feasibility:



- **Working Prototype:** NetAudit, a live React/Vite dashboard, already turns uploaded configs into compliance ratings using proven libraries (Netmiko, NAPALM).
- **Modular Vendor Support:** vendor logic lives in declarative dictionaries, not hard-coded parsers a new vendor means a new entry, not a rewrite.
- **Scalable by Design:** this architecture is what makes “any vendor” achievable rather than aspirational.

Economic Feasibility:



- **Open-Source Stack:** built entirely on open-source AI/ML and parsing libraries, with no per-vendor licensing fees.
- **No Vendor Lock-In:** independence from any single vendor's proprietary management suite keeps costs predictable.
- **Low Total Cost:** lower upfront and recurring costs than vendor-specific compliance tooling.

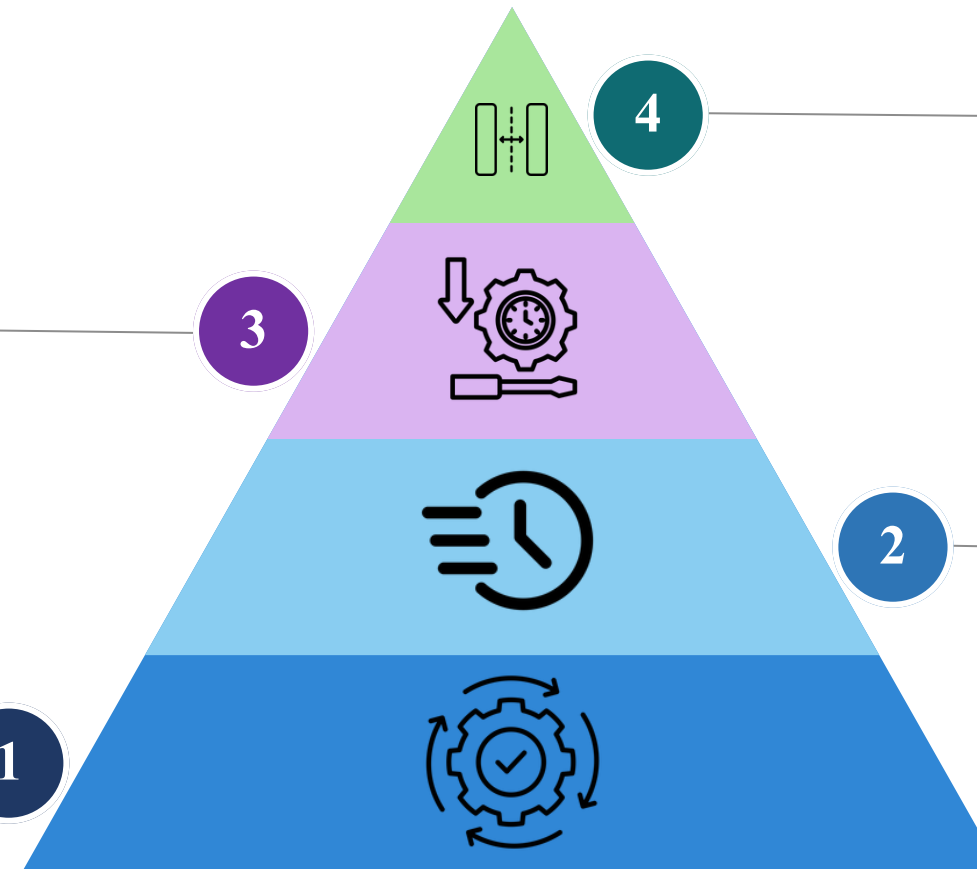
Impact

3. Frees Staff for High-Value Work

Removes repetitive checklist work from engineer's plates, freeing them to focus on real remediation and higher-value security tasks instead.

1. Consistent Compliance Posture

Delivers an always-current compliance picture across firewalls, routers, and switches from dozens of vendors vital for defence-grade infrastructure like NTRO's networks.



4. Closes the Misconfiguration Gap

Misconfiguration is one of the most common root causes of network breaches. NetAudit catches it between audit cycles before an attacker or auditor does.

2. Faster Audit Turnaround

Turns weeks of manual line-by-line config review against a benchmark PDF into a single upload a complete, evidence-backed report returned in minutes.

Benefits

01

Economic

Built entirely on open-source tools no per-vendor licensing cost or vendor lock-in. Savings compound through fewer audit hours and lower incident-response costs.

03

Operational

Coverage grows with every new configuration pattern encountered, without waiting on a vendor update. Every finding carries a full evidence trail to its source line.

02

Security

Enforces one hardening baseline CIS, NIST SP 800-53, DISA STIG, or ISO/IEC 27001 uniformly across every device, even hardware a vendor-locked tool wouldn't recognize.

04

Scalability

New vendor support is a lightweight dictionary entry, not a rewrite scaling cleanly to white-box switches, cloud-native groups, or hardware yet to exist.

1.	<u>NCIIPC (NTRO unit)</u>	• National Critical Information Infrastructure Protection Centre the nodal agency under NTRO for securing India's critical information infrastructure. Its guidelines directly shape the compliance mandate this tool is built to enforce. • https://www.nciipc.gov.in/
2.	<u>CIS Benchmarks</u>	• Consensus-built, vendor-specific secure configuration checklists published by the Center for Internet Security. Source of the actual rules (disable Telnet, no default SNMP strings, etc.) encoded into our Rego policies. • https://www.cisecurity.org/cis-benchmarks
3.	<u>NIST SP 800-53</u>	• NIST's catalog of security and privacy controls for federal information systems. Used to cross-reference control families access control, audit, configuration management — our engine checks against. • https://csrc.nist.gov/pubs/sp/800/53/r5/upd1/final
4.	<u>DISA STIGs</u>	• Security Technical Implementation Guides from the U.S. Defense Information Systems Agency. Provide hardened, defense-grade configuration baselines for high-security network deployments. • https://public.cyber.mil/stigs/
5.	<u>ISO/IEC 27001</u>	• International standard for Information Security Management Systems (ISMS). Anchors our audit-trail and reporting approach to a globally recognized compliance framework. • https://www.iso.org/isoiec-27001-information-security.html
6.	<u>Tenable Audit Files (reference)</u>	• Pre-built compliance audit files used by Tenable's vulnerability-management tools as an industry reference implementation. Used to validate that our Rego rule translations match real-world commercial audit tooling. • https://www.tenable.com/audits
7.	<u>NIST OSCAL</u>	• Open Security Controls Assessment Language a machine-readable format for expressing controls, assessments, and results. Informs the standardized JSON schema we use to represent compliance results across vendors. • https://pages.nist.gov/OSCAL/